Alberto Anelli

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EDUCATION

Sant'Anna School of Advanced Studies, Pisa
2025 – 2027

Honors Student – Second-Level Diploma (equivalent to a second-level master's degree) in Robotics Science and Technology – EQF 8 – Expected Oct 2027

University of Pisa
2025 – 2027

MSc in Bionics Engineering – Biorobotics track – EQF 7 – Expected Oct 2027

University of Pavia
2022 – 2025

BSc in Computer and Electronic Engineering – EQF 6
110/110 **cum laude**

IUSS Pavia – School for Advanced Studies
2022 – 2026

Honors Student – Diploma in Science and Technology
100/100 **cum laude**

Almo Collegio Borromeo, Pavia
2022 – 2025

Student at Collegio Borromeo, the oldest merit-based college in Italy.

Bari Conservatory of Music
2013 – 2018

Classical Piano Student

RESEARCH ACTIVITIES

Intern Research Engineer – SSSA – Medical Robotics
2026 – Present

Sant'Anna School of Advanced Studies – Pisa, Italy – Supervisor: **Prof. V. Iacovacci**

- Study of local magnetic control strategies using miniaturized coils mounted on an endoscopic device

Intern Research Engineer – MPI-IS – Medical Robotics
Jun 2025 – Sep 2025

Max Planck Institute for Intelligent Systems – Stuttgart, Germany – Supervisor: **Dr. sc. Jelena Zinnanti**

- Robotically assisted, on-demand hyperthermia for tissue ablation
- Hydraulically actuated soft catheter for interventional procedures under MRI

Undergrad Thesis Researcher – UNIPV – Control Theory
2024 – 2025

Prof. A. Ferrara, Prof. G.P. Incremona (POLIMI), Pavia, Italy

- Event-Triggered Eco-Driving for Vehicle Platoons in Urban Traffic Network

Undergrad Thesis Researcher – IUSS – Machine Learning
2024 – 2026

Prof. Taramelli Group, IUSS Pavia, Italy

- Web-enabled, user-friendly Python pipeline for rare earth element mapping from hyperspectral imagery

Research Intern – CERN Beamline for Schools
2021 – 2023

CERN BL4S Competition – DESY, Germany & INFN Bari, Italy

- DESY **Testbeam**, Germany: performed the ExTRa experiment on transition radiation in the beamline
- INFN, Bari, Italy: analyzed the ExTRa experiment data using Python

PUBLICATIONS

- *Energy-efficient control of vehicle platoons in urban traffic via event-triggered eco-driving*, submitted to ECC 2027 (25th European Control Conference), Brussels, Belgium **(First author)**
- *AI on hyperspectral satellite data for rare earth element detection: enabling sustainable resource management in the perspective of current and future satellite missions*, IGARSS 2026 (2026 IEEE International Geoscience and Remote Sensing Symposium), Washington, D.C.
- *The EXTRA-BL4S experiment for the measurement of the energy and angular distributions of transition radiation X-rays*, Journal of Instrumentation 18 (2023) P04017, DOI 10.1088/1748-0221/18/04/P04017

EXPERIENCES

1st Prize – Robotics and AI Hackathon
Oct 2026

Siemens Healthineers, Erlangen, Germany

- **Autonomous robotic navigation** of guidewires and catheters for **endovascular procedures**: developed a parallelized physics simulator to train privileged **reinforcement learning experts (SAC)**, whose rollouts

were distilled into a vision-language-action policy (**SmolVLA**, 450M) that steers from fluoroscopy images and text instructions.

Teaching Assistant
2023 – 2026

University of Pavia, Italy
- Assisted in Calculus I and II, Mathematical Methods and Linear Algebra courses for engineering students.

Consultant, IT Area
Oct 2023 – Nov 2024

Jeco Pavia, Pavia, Italy
- WordPress, Power BI, Streamlit, Flutter.

Cybersecurity Challenger
2021

CyberChallenge.IT National Final, UniBa, Bari, Italy
- Network Security, Cryptography, Social Engineering, Web Application Security, Operating System Security, Incident Response and Forensics, Penetration Testing, Reverse Engineering

Mentee
2023 – 2025

LeadTheFuture Mentoring Program, Italy
- Participated in a mentoring program designed to empower students in STEM with guidance from high-impact mentors.

CONFERENCES & SCHOOLS

Participant
Sep 2026

Selected participant in the Boston Study Tour Program

Speaker – Honorable Mention
Mar 2026

4th EELISA Scientific Student Competition, Madrid, Spain
- Delivered a presentation titled "*From Spectral Physics to AI: A Hybrid Approach for Reliable Rare Earth Element Detection in Hyperspectral Imagery*"

Participant
Aug 2024

Cornell, Maryland, Max Planck Pre-doctoral Research School in Computer Science, Saarbrücken, Germany

Fellow
Oct 2023

Selected Fellow of the Silicon Valley Fellowship Program

Speaker
Jun 2022

10th Beam Telescopes and Test Beams Workshop, Lecce, Italy
- Delivered a presentation titled "*A BL4S Experiment: EXTRA – Electron X-ray Transition RAdiation*"

HONORS & SCHOLARSHIPS

- Selected through two competitive national selection processes for the IUSS and Sant'Anna Honors Programs and awarded **full scholarships** for each (2022, 2025)
- Winner of the CERN Beamline for Schools (BL4S) Competition (2021)
- Awarded a €49,600 Erasmus Mundus Scholarship to attend the CODAS Master's Program at UPC Barcelona and Aalto University (1% admission rate) (Declined in favour of the SantAnna Honours Programme)
- Bronze and Gold Medal, Problem Solving Team Olympiad (2019, 2022)

SKILLS

Programming: Python, C/C++, MATLAB, Java, Dart, SQL, VHDL, Assembly
Robotics & Simulation: ROS2, LabVIEW, COMSOL, ANSYS, SOLIDWORKS
Tools & Other: ROOT, Streamlit, Flutter, Kali Linux
Piloting: EU UAS Open Category A1/A3 drone pilot certificate
Languages: Italian (Native), English (C1 CEFR), French (DELF B1)

OTHER EXPERIENCES

- IUSS Student Representative in RIASISSU (2024 – 2025)
- Choral Singer, Almo Collegio Borromeo Choir (2022 – 2025)
- Optimist and 29er Skipper, Competitive Sailor (2014 – 2020)